Study on Microstructure Evolution of Directionally Solidified Nickel-based Superalloy CM247LC [1]

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Abstract: A CM247LC Ni-based superalloy specimen was directionally solidified by the Bridgman method at a withdrawal rate of 2.5 mm/min to investigate the microstructural evolution along the solidification direction. The as-cast specimen was characterized by stereomicroscopy (SM), optical microscopy (OM), scanning electron microscopy (SEM), and energy dispersive spectroscopy (EDS), in conjunction with thermodynamic calculations performed in JMatPro and casting simulations performed in ProCAST. The results show that competitive growth of columnar grains occurs predominantly within 60 mm above the water-cooled chill plate, and that the variations in the primary and secondary dendrite arm spacings are governed mainly by the axial temperature gradient and the local growth rate. As the distance from the chill plate increases, the dendrite arm spacings continue to grow but at a progressively decreasing rate. The size evolution of as-cast microstructural features—including γ dendrites, γ/γ′ eutectics, and MC carbides—follows a trend similar to that of the dendrite arm spacings.

Keywords: Nickel-Based Superalloys, Directional Solidification, Competitive Growth, Microstructure, CM247LC

Nickel-based superalloys are widely used in aviation, aerospace, the nuclear industry, petrochemicals, and transportation owing to their excellent high-temperature strength, creep resistance, and oxidation and corrosion resistance [1-2]. Building upon conventional equiaxed casting, directional solidification eliminates transverse grain boundaries perpendicular to the loading direction and produces parallel columnar grains, thereby further improving the mechanical properties of superalloys [3]. The properties of directionally solidified superalloy castings are closely related to their microstructure, and controlling and optimizing the microstructure is an effective route to enhanced performance; systematic investigation of the microstructural evolution of directionally solidified columnar grains is therefore essential [4]. Numerical simulation can faithfully reproduce the microstructural evolution process and quantitatively establish the relationship between processing parameters and the resulting microstructure [5-6].

Szeliga et al. [7] successfully simulated the directional solidification of columnar grains in the nickel-based superalloy CMSX-4 using the CAFE module of ProCAST. Burbelko et al. [8] employed the same module to study the dendrite size of steel ingots and obtained results in good agreement with experimental data. Torroba et al. [9] used the CAFE module of ProCAST to simulate the grain morphology of nozzle guide vanes, providing strong support for research in this area. Hajkowski et al. [10] performed multiscale simulations of the columnar-to-equiaxed transition in the AlSi7Mg alloy with the CAFE and phase-field modules of Calcosoft and validated the predictions experimentally, thereby deepening the understanding of crystal-growth mechanisms. Gerstgrasser et al. [11] applied a cellular automaton model to simulate crystal growth during selective laser cladding, offering new insights for the development of laser processing technologies. Shang et al. [12] compared in-situ temperature profiles measured by a furnace temperature tracer during directional solidification with temperature-field predictions from ProCAST, and the good agreement between the two further confirmed the validity and accuracy of the simulation approach.

Although extensive work has been carried out on the numerical simulation of alloy solidification, progress has been limited by the difficulty of bridging micro- and macro-scales within existing solidification algorithms. It is therefore of considerable significance to combine numerical simulation with experiment in order to investigate the microstructural evolution during directional solidification. In the present work, directional solidification experiments were performed on the low-carbon nickel-based superalloy CM247LC, and, with the aid of simulation tools, the evolution of columnar grains, dendrite arm spacings, and other microstructural features at different heights along the solidification direction of the test bar was analyzed systematically.

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1. Materials and Methods

The alloy used in this study was CM247LC, with a nominal composition (mass fraction, %) of Ni–8.1Cr–5.6Al–1.0Ti–9.2Co–0.007Zr–0.015B–0.07C–3.2Ta–9.5W–0.5Mo. The alloy was melted in an ALD VIM-IC 02/100 vacuum induction melting and directional solidification furnace and poured into ceramic shell molds. Directional solidification was then carried out at a withdrawal rate of 2.5 mm/min, producing test bars in groups of eight, each with a diameter of φ20 mm and a length of 180 mm .

Longitudinal and transverse sections were cut from the test bars, and metallographic specimens were prepared by mounting, grinding, polishing, and etching. The etchant consisted of 20 mL HCl + 20 mL H_2O + 4 g $CuSO_4·5H_2O$. The microstructures of both sections were examined using a Zeiss Stemi 508 stereomicroscope (SM), a Zeiss optical microscope (OM), and a Nova NanoSEM 450 field-emission scanning electron microscope (SEM) equipped with an energy-dispersive X-ray spectrometer (EDS).

Because the specimens are three-dimensional whereas a cross-sectional micrograph captures only the two-dimensional morphology of the cut surface, several sub-regions sharing the same crystallographic orientation and enclosed by a single continuous boundary were counted as a single grain in this study.

The primary dendrite arm spacing on the transverse section was determined from

$$\lambda_1=(A/n_1)^{0.5} \quad (1)$$

where A is the analyzed cross-sectional area and n_1 is the number of cruciform dendrite cores within that area.

The secondary dendrite arm spacing on the longitudinal section was determined from

$$\lambda_2=L/n_2 \quad (2)$$

where L is the length of the selected primary dendrite trunk and n_2 is the number of secondary dendrite arms along L.

A finite-element model of the directional solidification process was constructed in ProCAST. After the geometric model was built and meshed, the composition of CM247LC was entered and the thermophysical properties were computed using the built-in thermodynamic database. The shell mold was defined as refractory alumina, the chill plate as copper, and the thermophysical properties of the other components were taken from the ProCAST material database. The interfacial heat-transfer coefficient between the shell mold and the alloy was set to 200 $W·m^{-2}·K^{-1}$, and that between the water-cooled chill plate and the alloy to 400 $W·m^{-2}·K^{-1}$. The hot-zone temperature was set to 1560 °C and the cold-zone temperature to 120 °C.

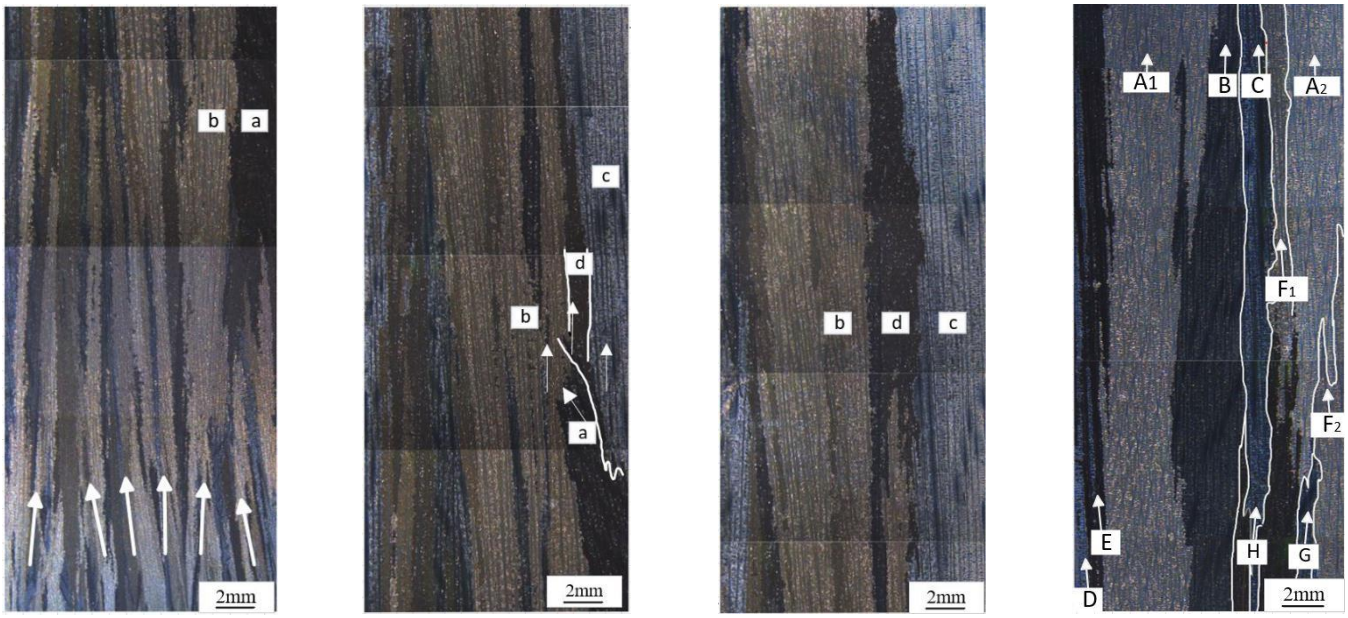
Non-equilibrium solidification of CM247LC was further simulated with the Scheil–Gulliver model in JMatPro V6.2 to obtain the elemental partitioning behavior and the evolution of phase formation as a function of temperature and residual liquid fraction. In addition, EDS analyses were performed at 60 micro-regions distributed along the axis of the as-cast specimen at a uniform spacing of 3 mm; each analyzed region measured 500 μm (transverse) × 500 μm (longitudinal).

2 Experimental Results and Analysis

2.1 Grain Structure

2.1.1 Test Results

Figure 1 shows stereomicrographs of the longitudinal section of the directionally solidified CM247LC casting after etching. The macrographs of the lower part of the casting (close to the water-cooled chill plate, i.e. the leading edge of solidification; each segment is 30 mm long) clearly reveal intense competitive growth among columnar grains of various orientations during the initial stage of directional solidification, as shown in Figs. 1a–1c. The macrograph of the upper part of the casting (far from the chill plate, i.e. the end of solidification; 30 mm long) shows that the columnar grains exhibit a well-developed ⟨100⟩ preferred orientation with only a small deviation from the heat-flow direction, as shown in Figure. 1d. Only a limited number of grains survived at the solidification front, and their boundaries are jagged and irregular. Some grains, such as A1, A2, F1, and F2 in Figure. 1d, appear as fully enclosed closed-loop boundaries on the section; this is an artefact of the two-dimensional section, which cannot fully represent the three-dimensional grain morphology.



(a) 1–30 mm (b) 30–60 mm (c) 60–90 mm (d) 150–180 mm

Figure. 1 Grain morphology of longitudinal section

Figure 2 shows the variation in the number and width of columnar grains along the longitudinal section of the test bar. The number of columnar grains decreases rapidly within the 1–60 mm range above the chill plate, after which the rate of change slows markedly. A comparison of the average- and median-width curves in Figure. 2 indicates that the two values differ little between 1 and 30 mm, implying that the columnar grains are relatively uniform in size during the initial stage of growth. Above 30 mm, however, competitive growth causes the grain sizes to become non-uniform: the median width drops below the average, and the macrostructure consists of a few large grains together with many smaller ones. Once the columnar grains have grown beyond 60 mm—and up to the completion of solidification—both the number and the size of the grains change only slightly. Overall, the dominant grain-competition stage occurs between 1 and 60 mm in height, during which most grains are eliminated and the grain count is substantially reduced.

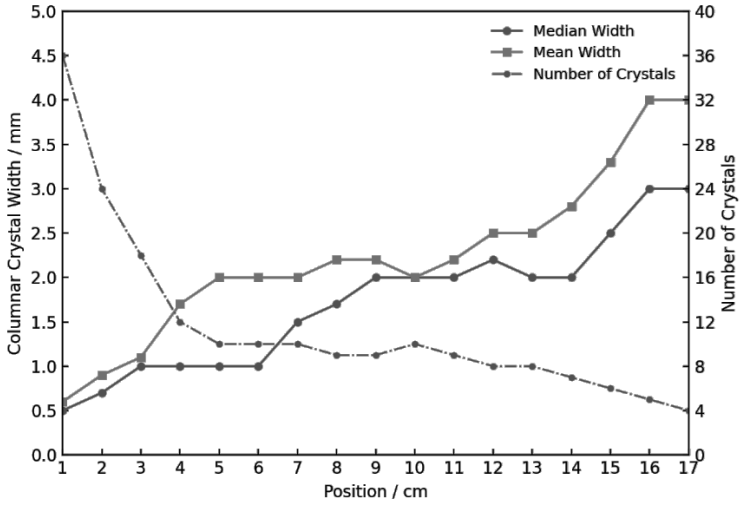


Figure. 2 Trend of the number and size of columnar grains in the longitudinal section

2.1.2 Analysis and Discussion

As shown in the lower part of Figure. 1a, the quenching effect imposed by the water-cooled copper chill plate produces a large number of densely packed grains during the initial stage of solidification. These columnar grains, exhibiting varying thicknesses and preferred orientations, are randomly distributed and subsequently grow rapidly along their respective preferred orientations. Most of the columnar grains display a certain orientation deviation angle with respect to the direction opposite to the heat flux. During competitive growth, grains with larger orientation deviation angles are progressively eliminated by those whose preferred orientation is more closely aligned with the direction opposite to the heat flux, whereas

grains well aligned with this direction continue to grow. In Figure. 1a, both grain a and grain b are well developed; in Figure. 1b, however, grain a converges with grain b while diverging from grain c, leading to the eventual elimination of grain a and the emergence of grain d at the intersection of grains a and c. At a greater distance from the chill plate, as shown in Figure. 1c, grains b, c, and d, all of which possess favorable preferred orientations, occupy the majority of the casting cross-section.

The essence of grain competition lies in the formation of dendritic branches and the migration of grain boundaries, both of which depend on the relative orientation of the grains and the local solute diffusion field [7–8]. When two grains converge, the solute fields at the dendrite tips overlap, causing the solute boundary layer to expand and thereby slowing the migration rate of solute elements across the solid–liquid interface [9–10]. In Figure. 1b, grain b exhibits a smaller orientation deviation angle relative to the heat-flux direction than grain a and therefore grows ahead. The diffusion field associated with grain b suppresses the leftward growth of grain a, while grain b itself continues to advance: tertiary dendrites develop on its secondary dendrite arms and subsequently evolve into dendrite trunks parallel to the primary dendrite, driving the grain boundary to shift toward the right. Previous studies have also reported that the orientation of secondary dendrite arms does not affect the outcome of competitive growth [17]. When two adjacent dendrites gradually separate owing to different growth directions, a blank region unaffected by the solute field forms at the divergent grain boundary [15]. In Figure. 1b, grains a and d exhibit such a divergent relationship as a result of a large orientation deviation. Grain d may either be a newly nucleated grain or another dendrite that has grown forward from the rear in three-dimensional space.

Figure 1d shows the late stage of solidification. At this stage, the number of grains observed on the cross-section is markedly reduced, and both the preferred orientations of the dendrites and their deviation angles from the heat-flux direction are small. Within the 30 mm length shown in Figure. 1d, only grains G and H are eliminated, whereas the remaining six grains (A–F) survive. The expansion and contraction of the grains at this stage are still governed by the direction of the heat flux and display a degree of randomness. When a casting undergoes directional solidification within a set of multiple, annularly arranged modules, direct radiative heat transfer toward the inner side is difficult to achieve. Moreover, during the directional solidification of these modules, the various solidification parameters vary with both time and space, and the heat-flux direction is not entirely parallel to the withdrawal direction. This may explain why grains D and E, despite their relatively small cross-sectional areas, are able to persist throughout the solidification process [18].

2.2 Dendritic Microstructure

2.2.1 Experimental Results

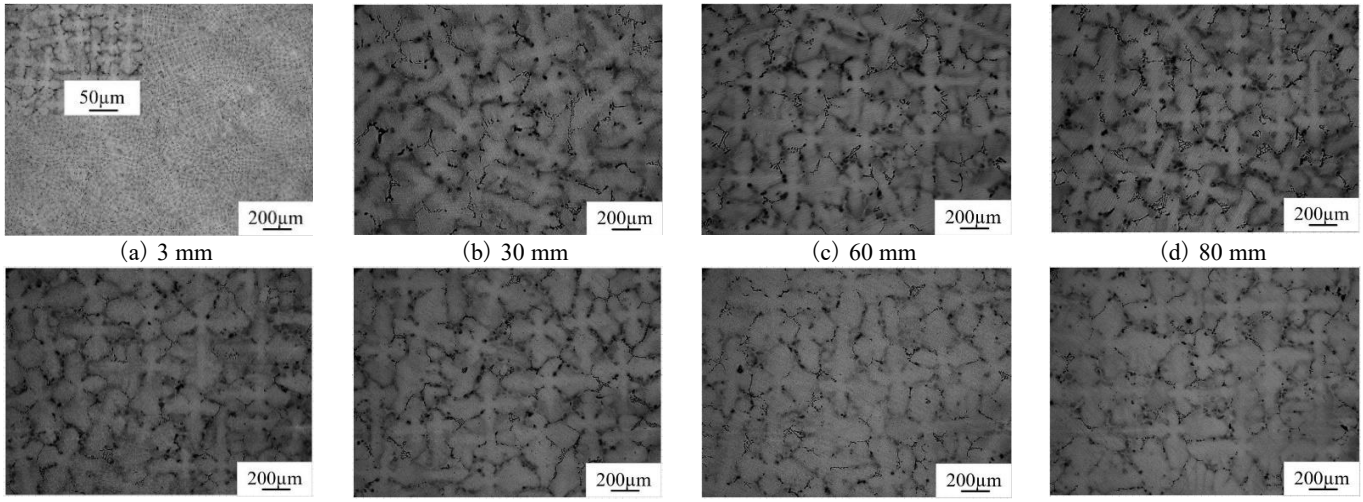
Figure 3 shows OM images of the transverse microstructure of the directionally solidified CM247LC alloy. The transverse dendrites exhibit a two-dimensional cruciform morphology. The dendrites in the lower region are relatively fine, whereas those in the upper region are coarser. Within certain areas, the arrangement becomes regularized, and cruciform dendrites sharing the same orientation over a localized range constitute individual columnar grains. As shown in Figure. 3a, the primary dendritic structure at a height of 3 mm above the chill plate is extremely dense; in contrast, at heights of 30 mm and above, no significant differences are observed and the dendrite sizes become relatively uniform.

Figure 4 shows OM images of the longitudinal microstructure of the directionally solidified CM247LC alloy. The longitudinal dendrites exhibit a continuous, parallel growth pattern oriented at a small angle to the withdrawal direction, and the orientation differences among the secondary dendrites give rise to variations in the thickness of the dendrite arms observed on the two-dimensional section. A combined examination of Figs. 3 and 4 reveals that competitive growth takes place between dendrites of both similar and different orientations, and that tertiary dendrite arms emerge at heights as low as 60 mm. This observation is consistent with the mechanism discussed earlier, namely that grain-boundary migration during competitive growth relies on dendritic branching.

Three regions of 1.5 × 1.5 mm were selected on both the transverse and longitudinal sections at different heights along the test bar, and the corresponding primary and secondary dendrite arm spacings were measured using Eqs. (1) and (2); the results are presented in Figure. 5. A combined analysis of Figs. 3–5 indicates that near the chill plate the dendritic structure is relatively dense and both the primary and secondary dendrite arm spacings are significantly smaller. As the distance from the chill plate increases, the dendrite arm spacings increase rapidly between 0 and 30 mm, rise only slightly between 30 and 60 mm, and remain essentially unchanged at heights of 60 mm and above. The primary dendrite arm spacing fluctuates around

450 μm, whereas the secondary dendrite arm spacing fluctuates around 100 μm.

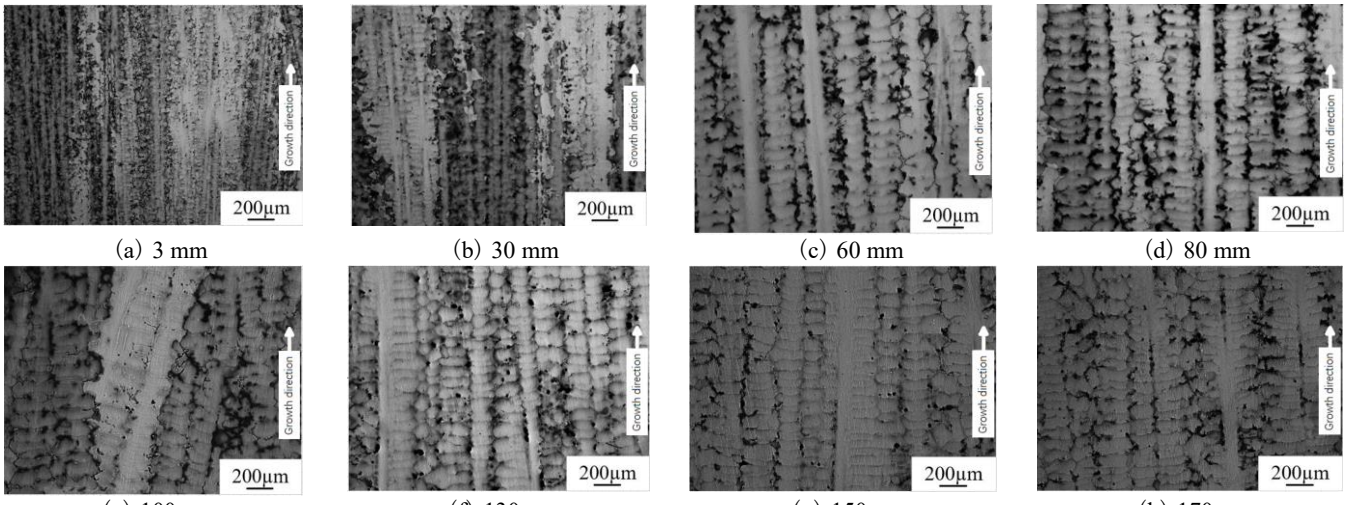
As shown in Figure. 3c, perturbations develop on the surfaces of the secondary dendrite arms at a height of 60 mm, leading to the formation of tertiary dendrite arms. As dendritic growth proceeds, the number of grains decreases markedly as a result of competitive growth, while the proportion of dendrites bearing tertiary arms gradually increases. In Figure. 3f, dendrites with tertiary branches already occupy approximately half of the cross-sectional area.



(a) 3 mm (b) 30 mm (c) 60 mm (d) 80 mm

(e) 100 mm (f) 130 mm (g) 150 mm (h) 170 mm

Figure. 3 Cross-sectional dendritic morphology for different heights from the water-cooled chill plate



(a) 3 mm (b) 30 mm (c) 60 mm (d) 80 mm

(e) 100 mm (f) 130 mm (g) 150 mm (h) 170 mm

Figure. 4 Longitudinal dendritic morphology for different heights from the water-cooled chill plate

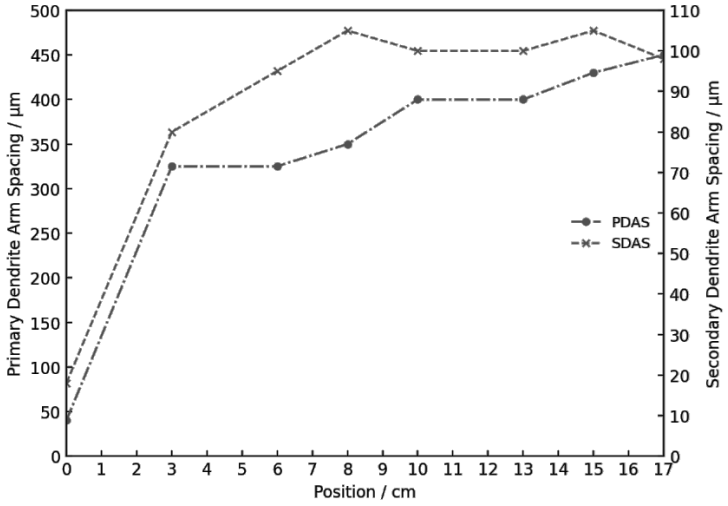


Figure. 5 Dendrite-arm spacing changing with distance from water-cooled chill plate

2.2.2 Analysis and Discussion

KURZ et al. [19] employed a critical stability theory model to express the relationship between the primary dendrite arm spacing and the temperature gradient and growth rate during directional solidification as follows:

$$\lambda_1 \propto G^{-1/2} \cdot R^{-1/4} \tag{3}$$

The FURER–WUNDERLIN model [20], which is based on the secondary dendrite coarsening hypothesis, expresses the relationship between the secondary dendrite arm spacing and the temperature gradient and growth rate during directional solidification as:

$$\lambda_2 \propto (G \cdot R)^{-1/3} \tag{4}$$

where G is the temperature gradient and R is the growth rate. That is, both the primary and secondary dendrite arm spacings increase as the temperature gradient and growth rate decrease. As shown in Figs. 3 and 4, the primary and secondary dendrite arm spacings in the test bar microstructure increase markedly with increasing height above the chill plate.

The numerical simulations yield cooling curves at 36 equidistant points spaced 5 mm apart along the specimen axis. Subsequent data processing provides the simulated values of the solidification-front temperature gradient G_i ($i \in [1, 36]$) and the columnar-grain growth rate R_i. The solidification-front temperature gradient G_i at the i-th point is defined as the ratio of the temperature difference between the (i+1)-th and i-th points to the 5 mm distance between them, evaluated at the moment when the i-th point cools to the liquidus temperature:

$$G_i \approx \frac{\triangle T}{\triangle Z} = \frac{T_{i+1} - T_i}{5} \tag{6}$$

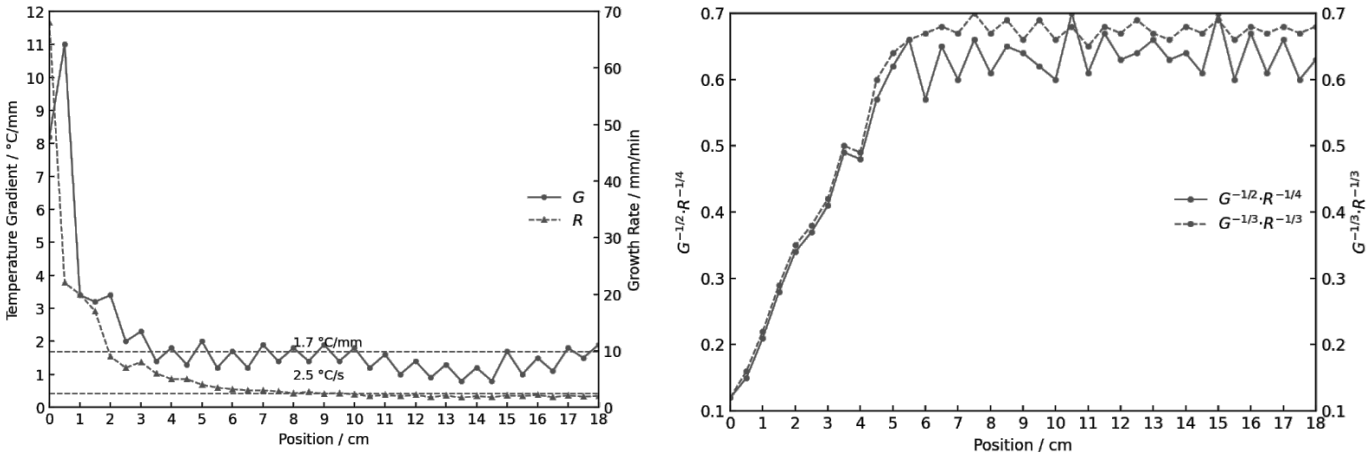
where， T_{i+1} is the temperature at point (i+1) at the instant when $T_i = T_L$, in °C.

Defining the columnar-grain growth rate R as the migration rate of the liquidus isotherm, the growth rate at the i-th point is given by:

$$R_i \approx \frac{\triangle Z}{\triangle t} = \frac{5}{t_{i+1} - t_i} \tag{7}$$

where t_i is the time at which T_i = T_L (in minutes) and t_{i+1} is the time at which T_{i+1} = T_L (in minutes).

Figure 6 shows the simulated solidification parameters along the test bar.



(a) Temperature gradient and growth rate (b) Calculated values from the theoretical model

Figure. 6 Simulation results based on solidification parameters

A comparison of the measured dendrite arm spacings in Figure. 5 with the theoretical values calculated from the simulated temperature field in Figure. 6a using Eqs. (3) and (4), as plotted in Figure. 6b, shows that the variation in the primary dendrite arm spacing is in good agreement with that of $G^{-1/2} \cdot R^{-1/4}$, while the variation in the secondary dendrite arm spacing is also strongly correlated with that of $(G \cdot R)^{-1/3}$.These results indicate that the temperature gradient and growth rate at the solid–liquid interface are directly related to the dendrite arm spacing in directionally solidified castings. During solidification by the Bridgman method, the solid–liquid interface gradually moves away from the chill plate, and the cooling effect provided by the plate progressively weakens as heat conduction becomes less efficient; heat extraction is then accomplished mainly through radiation from the shell mold, which leads to a reduction in the temperature gradient ahead of the solid–liquid interface. Consequently, the dendrite arm spacings in Figs. 3 and 4 first increase progressively and subsequently remain essentially constant.

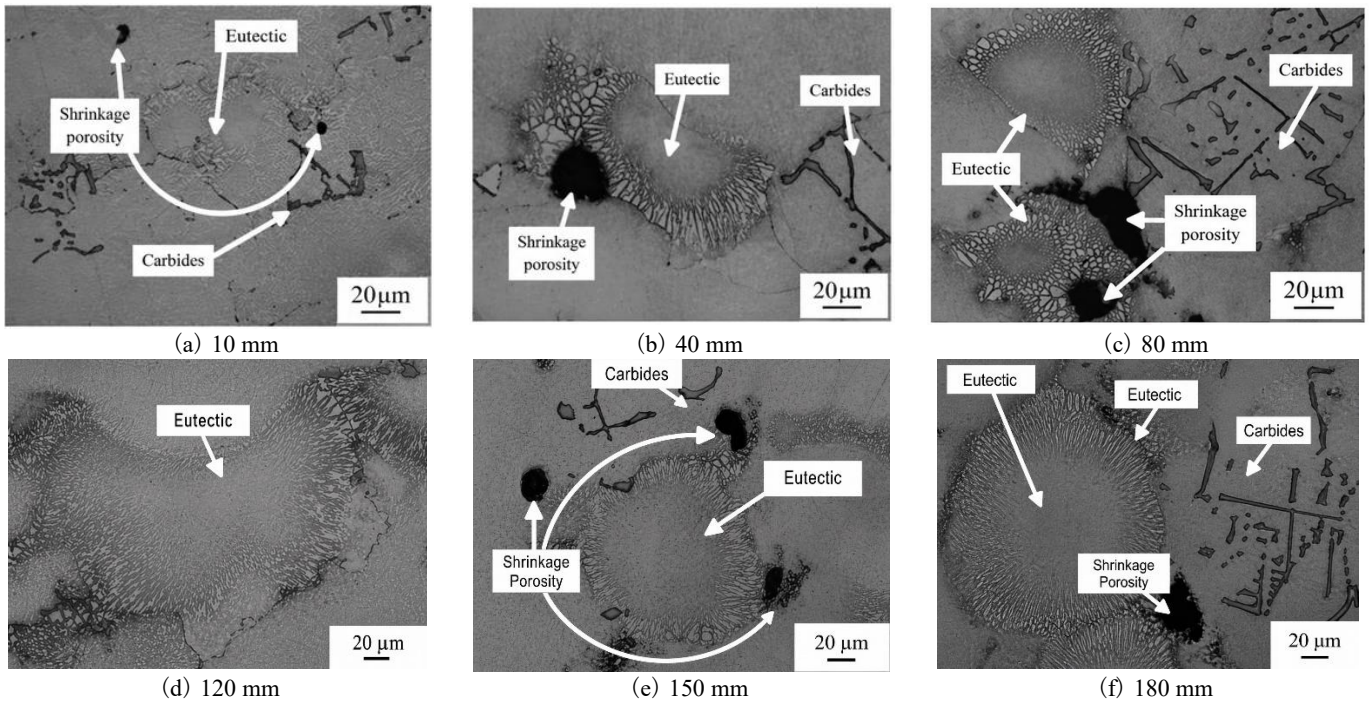
2.3 Microstructure

2.3.1 Experimental Results

Figure 7 shows the microstructure of the CM247LC alloy. The as-cast microstructure contains a certain amount of γ/γ′ eutectic and carbides. It is generally accepted that eutectic phases and carbides tend to precipitate in the interdendritic regions and are seldom observed within the dendrite trunks, which is governed by compositional microsegregation and the characteristics of dendritic growth [24]. As shown in Figure. 7, the γ/γ′ eutectic exhibits a radial morphology, and shrinkage cavities are frequently observed in the vicinity of the eutectic regions. With increasing height above the chill plate, both the number and the size of the eutectic regions show a tendency to increase progressively. The eutectic regions in Figure. 7a are markedly smaller than those at higher positions, and the morphology of the eutectic clusters evolves from a typical semicircular shape in Figs. 7b–7d to a circular shape in Figs. 7e and 7f.

Owing to the addition of a small amount of carbon to CM247LC, the as-cast microstructure consists not only of γ-phase dendrites and γ/γ′ eutectic clusters but also of primary carbides (MC-type carbides). In Figs. 7a–7f, the carbides exhibit a script-like morphology. At a height of 10 mm, the carbides are relatively dispersed, whereas at heights above 30 mm they appear predominantly as continuous script-like structures with no significant variation in size.

Porosity is primarily distributed at the edges of the eutectic regions and within the interdendritic spaces, and appears mainly in circular, irregular elongated, or triangular shapes. It is generally accepted that the formation of porosity in nickel-based superalloys is related to the temperature gradient of the liquid phase ahead of the solid–liquid interface and to the withdrawal rate during processing, with the pore size decreasing continuously as the withdrawal rate increases. The results of the present experiment show that the size of the shrinkage pores increases with the distance from the chill plate only within the initial solidification zone (0–40 mm); beyond this range, the size of the shrinkage pores remains essentially unchanged.



(a) 10 mm (b) 40 mm (c) 80 mm

(d) 120 mm (e) 150 mm (f) 180 mm

Figure. 7 Microstructure between dendrite arms at different distances from the water-cooled chill plate

2.3.2 Analysis and Discussion

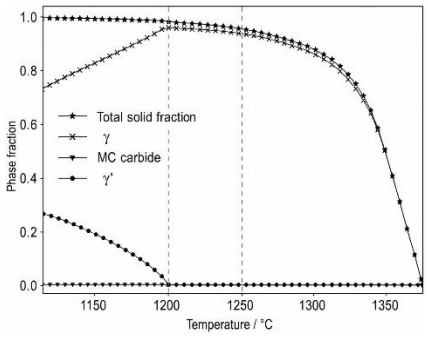
Figure 8 shows the thermodynamic calculation results for the CM247LC alloy. As shown in Figure. 8a and reported in [20–21], during the solidification of the nickel-based superalloy CM247LC the γ solid solution first precipitates from the liquid phase: $L_1 \rightarrow \gamma + L_2$. Subsequently, MC carbides form in the interdendritic regions: $L_2 \rightarrow \gamma + L_3 + MC$. As solidification approaches completion at 1255 °C, with the solid fraction reaching approximately 95%, the eutectic reactions $L_3 \rightarrow \gamma/\gamma'$ and $L_3 + \gamma \rightarrow \gamma/\gamma'$ take place. At 1210 °C the solid fraction reaches approximately 98%. As the temperature continues to decrease, the solubility of solute atoms in the γ solid solution diminishes, and a solid-state phase transformation occurs in the supersaturated γ matrix, in which the γ′ phase nucleates, precipitates, and grows: $\gamma \rightarrow \gamma'$.

The core of the γ/γ′ eutectic cluster forms instantaneously through the eutectic reaction and is followed by the rapid growth of the γ/γ′ phase, which gradually evolves into a coarse lamellar eutectic. As a result, the eutectic structure at the cluster core remains very fine, while the surrounding structure exhibits a coarse, divergent pattern, giving rise to the petal-like morphology.

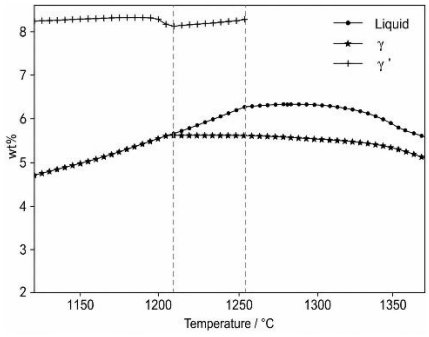
Furthermore, previous studies have shown that the contents of Co, Cr, and W gradually decrease from the center toward the periphery of the eutectic, whereas the contents of Al, Hf, Ta, and Ti gradually increase [27]. This compositional variation is generally consistent with that of the γ′ phase formed within the eutectic temperature range during solidification (1210–1255 °C); in contrast, the composition of the γ phase formed within this temperature range does not exhibit pronounced variation, as shown in Figs. 8b–8i. These results indicate that the thermodynamic calculations are reliable and that the simulation approach is valid.

Previous studies have suggested that the eutectic clusters in carbon-bearing nickel-based superalloys exhibit several characteristic morphologies, namely γ-based eutectics that nucleate on the γ phase, eutectics that nucleate on MC carbides, and free eutectics that nucleate directly from the liquid phase [23–24]. Among these, eutectic clusters that nucleate on a γ-phase substrate are approximately semicircular in shape, with the eutectic nucleus located at the tip of a γ dendrite and sharing the same crystallographic orientation as the γ substrate. For eutectic clusters that nucleate on MC carbides, the crystallographic orientation is determined by the carbide. Free γ/γ′ eutectic clusters that nucleate and grow directly from the liquid are approximately circular, with the eutectic nucleus at the center of the cluster and a random crystallographic orientation [28]. Eutectic growth continues until impingement with neighboring eutectic clusters or adjacent γ dendrites; consequently, the size and spacing of the γ dendrites place an upper bound on the size of the eutectic clusters. The eutectic structures vary in both size and shape, and become coarser with increasing distance from the chill plate. This is because, during the initial stage of solidification, the primary dendrite arm spacing is small and the dendritic structure is dense, which restricts the growth of the eutectics and thereby results in finer eutectic clusters.

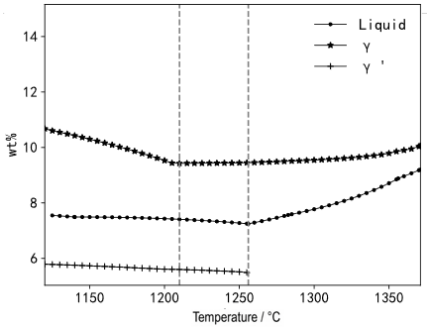
The directional solidification of superalloys proceeds from the bottom upward, and consequently macroscopic porosity is relatively scarce, with most porosity manifesting as interdendritic microporosity. As shown in Figs. 5 and 6, with increasing distance from the chill plate the dendrite arm spacings continue to broaden, the size of the interdendritic liquid pockets correspondingly increases, the as-solidified microstructure becomes progressively less uniform, and both the eutectic clusters and the shrinkage cavities increase markedly in size. This phenomenon arises because the interdendritic regions are the last to solidify during directional solidification. Within the mushy zone, dendrites interlock with one another and partition the residual liquid metal into numerous small liquid pockets; when these isolated pockets can no longer be replenished by external liquid feeding, interdendritic microporosity develops. Several studies have further proposed that, during the directional solidification of carbon-bearing nickel-based superalloys, the presence of carbides in the mushy zone markedly reduces its permeability, thereby impeding interdendritic feeding and aggravating the formation of shrinkage cavities [25–26]. This is consistent with the observation in Figure. 7 that shrinkage cavities adjacent to eutectic regions are frequently associated with carbides.



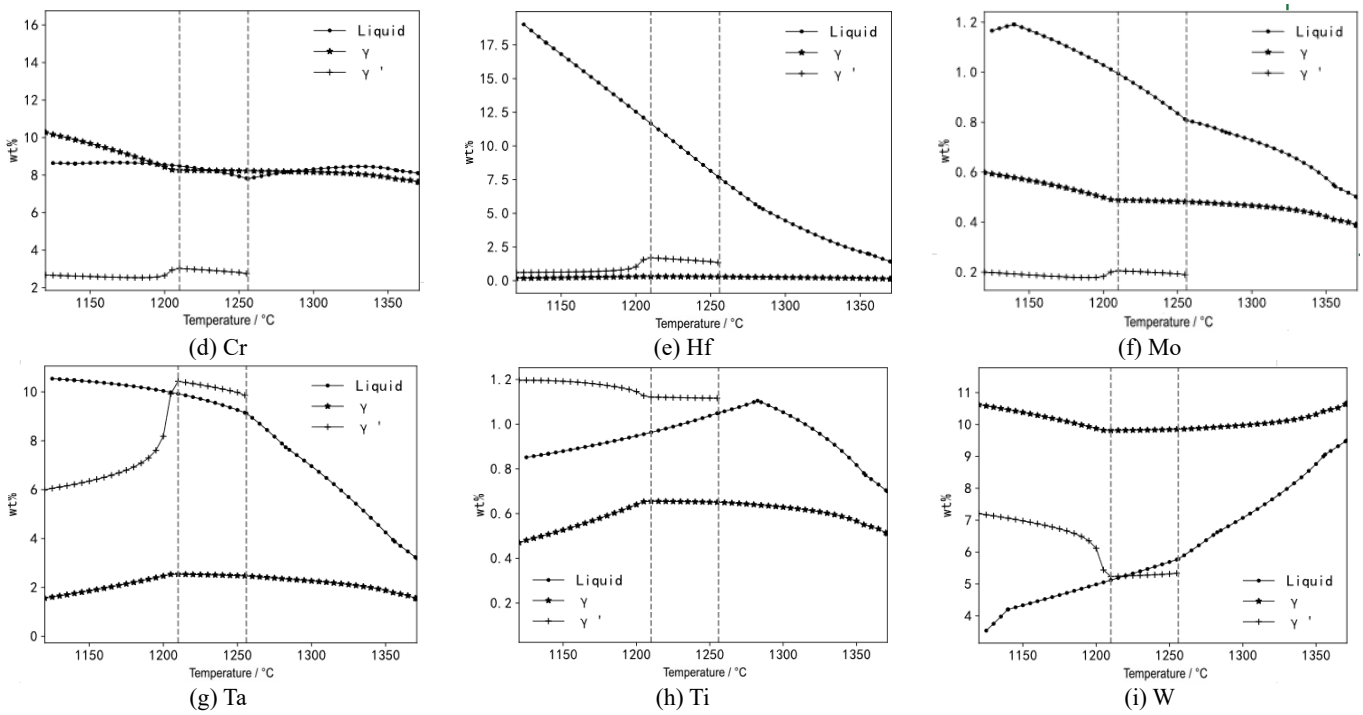
(a) Phase diagram



(b) Al



(c) Co



(d) Cr (e) Hf (f) Mo

(g) Ta (h) Ti (i) W

Figure. 8 Thermodynamic calculation results of CM247LC

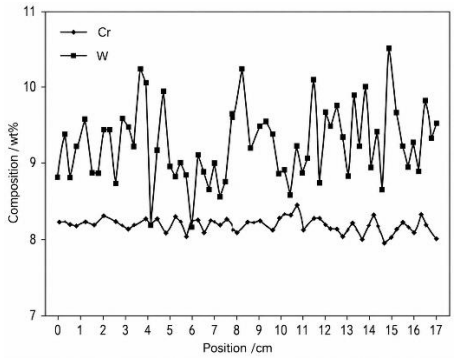
Element segregation during directional solidification occurs predominantly as microsegregation within and between dendrites, while compositional inhomogeneity is not evident on the macroscopic scale. It is therefore essential to investigate the microsegregation behavior of elements in nickel-based superalloys [32]. Figure 9 shows the composition distribution measured along the longitudinal section of the directionally solidified test bar, and Table 1 lists the segregation coefficients of the various elements in the superalloy. As shown in Figure. 9, the macroscopic composition measurements confirm that no significant macrosegregation develops along the growth direction of the nickel-based superalloy specimen during directional solidification.

A combined examination of Figure. 8 and Table 1 reveals that, as the volume fraction of the liquid phase decreases, the concentrations of positive segregation elements such as Al, Ti, Ta, and Hf ($k < 1$; Figs. 8b, 8e, 8g, and 8h) progressively increase in the residual liquid. These elements consequently enrich in the interdendritic regions and ultimately reach the eutectic composition, leading to the formation of the γ/γ′ eutectic phase shown in Figure. 7. In contrast, the contents of negative segregation elements such as W and Co ($k > 1$, Figs. 8c and 8i) decrease in the liquid, and these elements therefore segregate to the dendrite trunks. These results are generally consistent with previously reported experimental measurements [28–32]. During dendritic growth, the microsegregation arising from the differences in the segregation coefficients of the various elements exerts only a minor influence on the macroscopic composition distribution of the casting; at the microscopic scale, however, it gives rise to pronounced compositional fluctuations, i.e., dendritic segregation.

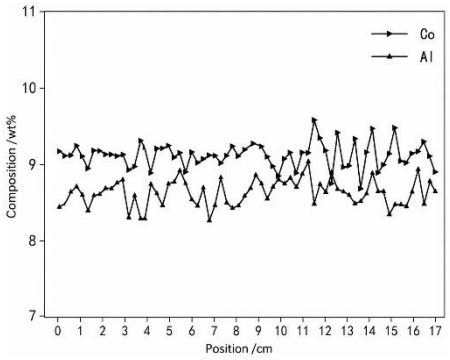
Figure 10 shows the standard deviation of the axial distributions of the various elements. It is worth noting that, despite the different segregation tendencies, the magnitude of compositional fluctuation along the macroscopic growth direction is approximately positively correlated with atomic number. This may be attributable to the solubility of the solute elements in the γ phase, or alternatively to the comparatively low diffusion coefficients of atoms with larger atomic radii in the liquid phase, which hinder homogeneous mixing [38].

Table 1 Segregation coefficients of various elements in the superalloy

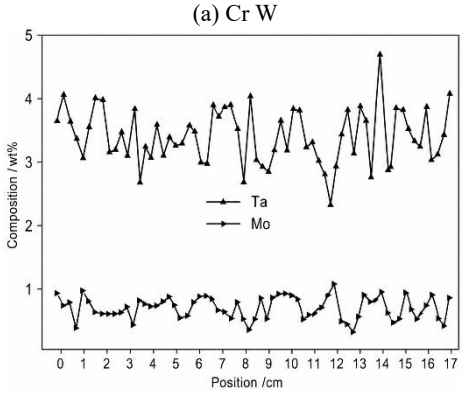
	Al	Ti	Cr	Co	Mo	Hf	Ta	W
HOBBS R A et al.[39]	0.91	-	1.05	1.06	-	-	0.76	1.47
Gao Shuang et al.[36]	0.91	0.88	0.97	1.12	0.86	-	0.54	-
Phase diagram calculation	0.89	0.67	1.04	1.27	0.62	0.18	0.35	1.64



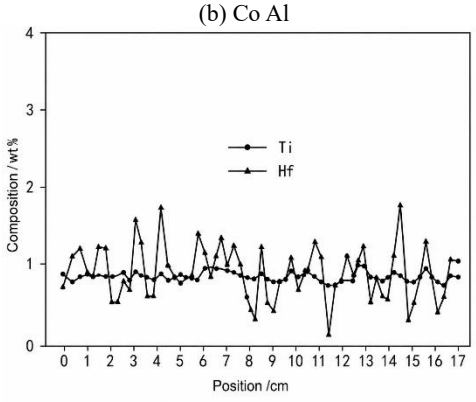
(a) Cr W



(b) Co Al



(c) Ta Mo



(d) Ti Hf

Figure. 9 Measured macro-inhomogeneity along the longitudinal section of a directionally solidified specimen

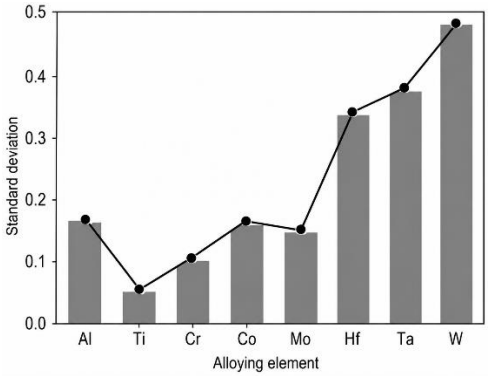


Figure. 10 Standard deviation of axial concentration distribution

3. Conclusions

(1) In the directionally solidified specimens of the CM247LC superalloy, the principal stage of competitive growth among columnar grains takes place at heights between 1 and 60 mm above the chill plate. During this stage, most grains are eliminated, leading to a marked reduction in the grain count. Above 60 mm, both the decrease in grain number and the increase in grain size proceed at a substantially slower rate.

(2) The as-cast microstructure of CM247LC consists predominantly of γ-phase dendrites. The dendrite arm spacings increase rapidly during the initial stage of solidification and the rate of increase progressively diminishes above a height of 60 mm. The variation in the dendrite arm spacings is in good agreement with the predictions of the temperature-field simulation.

(3) The as-cast microstructure of CM247LC also contains a certain amount of γ/γ′ eutectic and MC-type carbides, together with distinct microporosity. The size variations of the eutectic clusters, carbides, and microporosity follow trends generally consistent with those of the dendrite arm spacings.

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镍基高温合金CM247LC定向凝固组织的演化研究

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摘 要 采用Bridgman法以2.5 mm/min的抽拉速度制备高温合金CM247LC定向凝固试样，研究定向凝固高度方向上的组织演变。采用体视显微镜（SM）、金相显微镜（OM）、扫描电镜（SEM）、能谱仪（EDS）及JMatPro热力学计算软件、ProCAST铸造模拟软件等对铸态试样组织和性能进行表征和研究。结果表明，柱状晶粒的竞争生长主要发生于距水冷盘60 mm以下的高度范围内；枝晶臂间距的变化主要依赖于轴向温度梯度与生长速度的变化，随距水冷盘距离提升，枝晶臂间距增加的趋势变缓；CM247LC的铸态组织除γ枝晶外，还包括γ/γ′共晶和*M*C碳化物，其尺寸变化规律与枝晶臂间距一致。

关键词 镍基高温合金；定向凝固；竞争生长；显微组织；CM247LC
中图分类号 TG132.3+2；TG244+3 **文献标志码** DOI：10.15980/j.tzzz.2024.08.019

Microstructure Evolution of Directionally Solidified Ni-based CM247LC Superalloy

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Abstract: Ni-based CM247LC superalloy specimen was directionally solidified using Bridgman method with pulling speed of 2.5 mm/min to investigate the microstructure evolution along solidification direction. The microstructure and performance of as-cast specimen were characterized by stereomicroscope (SM), metallurgical microscope (OM), scanning electron microscope (SEM), energy spectrometer (EDS), as well as thermodynamic calculation software JMatPro and casting simulation software ProCAST. The results indciate that the competitive growth of columnar grains mainly occurs in the range of height below 60 mm from the water-cooling plate. The variation of dendrite arm spacing mainly depends on the axial temperature gradient and variation of growth rate, which gradually slows down with the distance from water-cooling plate increasing. The as-cast microstructure is composed of γ/γ' eutectic and *M*C carbides in addtition to γ dendrites, of which the size variation law is similar to that of dendrite arm spacing.

Key Words: **Ni-based Superalloys**, **Directional Solidification**, **Competitive Growth**, **Microstructure**, **CM247LC**

镍基高温合金因具有优异的高温强度、抗蠕变、抗氧化和耐腐蚀性能，被广泛应用于航空、航天、核工业、石油化工以及交通运输等领域[1-2]。在传统等轴晶铸造的基础上，通过引入定向凝固技术消除了载荷方向的横向晶界，发展为相互平行的柱状晶组织，进一步提高了高温合金的力学性能[3]。高温合金定向凝固铸件的性能与其显微组织紧密相关，对铸件的组织结构进行控制和优化是提升铸件性能的有效手段，因此有必要对定向凝固柱状晶的组织演变进行系统的研究[4]。

采用数值模拟技术能有效再现凝固组织演变的过程，定量确定工艺因素与凝固组织之间的关系[5-6]。SZELIGA D等[7]采用ProCAST软件的CAFE模块模拟了CMSX-4镍基合金柱状晶的定向凝固过程；BURBELKO A等[8]也采用该软件对钢碇的枝晶尺寸进行了仿真研究，发现其结果与试验结果高度一致；TORROBA A J等[9]则采用ProCAST的CAFE模块模拟了管嘴引导叶片的晶粒形貌。HAJKOWSKI J等[10]采用Calcosoft软件对AlSi7Mg合金由柱状晶向等轴晶的转变过程进行了多尺度模拟与验证，深化了对晶体生长机制的理解。GERSTGRASSER M等[11]则采用元胞自动机模型模拟了选区激光熔覆中的晶体生长行为，为激光加工技术的发展提供了新的思路。尚伟等[12]将炉温跟踪仪测得的定向凝固实际温度曲线与ProCAST温度场模拟结果进行了对比，验证了模拟方

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法的有效性和准确性。虽然对合金凝固过程的数值模拟做了大量工作，但是由于现有凝固模拟算法难以兼顾微观宏观尺度，限制了其发展，因而结合数值模拟与试验研究定向凝固的组织演变规律具有积极意义。本研究采用低碳镍基高温合金 CM247LC 进行定向凝固试验，配合模拟仿真工具，沿试棒凝固方向分析不同生长高度上柱状晶晶粒、枝晶间距及其他显微组织演变规律，旨在为镍基高温合金的应用提供参考。

1　试验材料与方法

试验合金为 CM247LC 合金，其成分为 Ni-8.1Cr-5.6Al-1.0Ti-9.2Co-0.007Zr-0.015B-0.07C-3.2Ta-9.5W-0.5Mo（质量分数）。采用 ALD-VIM-IC 5 E/DS/SC 真空定向凝固炉，将合金熔炼后浇注入陶瓷型壳中，以 2.5 mm/min 的抽拉速度进行定向凝固，制备出 8 件 1 组的 ϕ 20 mm×180 mm 的等截面试棒。

对试棒进行纵截面和横截面的剖切，经过镶嵌、研磨、抛光和腐蚀制得金相试样，腐蚀液采用配比为 20 mL 的 HCl＋20 mL H_2O＋4 g 的 $CuSO_4 \cdot 5H_2O$ 的混合溶液。采用 Stemi 508 体视显微镜（SM）、Zeiss 光学显微镜（OM）和 NovaNano450 场发射扫描电镜（SEM）对定向凝固试样的横截面和纵截面进行观察。因试样实际存在于三维空间，而剖面图仅能表现切面的二维形貌，本研究中晶粒计数时将存在多个封闭轨迹边界而取向一致的晶粒认定为同一个晶粒。

采用下式计算横截面上的一次枝晶臂间距 λ_1：

$$\lambda_1 = (A/n_1)^{0.5} \tag{1}$$

式中，A 为横截面面积；n_1 为横截面内十字花状枝晶干的数目。

采用下式计算纵截面上的二次枝晶臂间距 λ_2：

$$\lambda_2 = L/n_2 \tag{2}$$

式中，L 为选取的枝晶干长度；n_2 为长度 L 上的二次枝晶臂数目。

利用 ProCAST 软件构建定向凝固过程三维模型并划分网格，输入 CM247LC 合金成分后通过软件热力学数据库计算得到热物性参数，型壳所用材料为耐火 Al_2O_3，水冷盘所用材料为 Cu，其他构件热物性参数通过 ProCAST 软件数据库获取。型壳与合金、水冷盘与合金间传热系数设分别设为 200、400 W/(m^2·K)；热区温度设为 1 560 ℃；冷区温度设为 120 ℃。

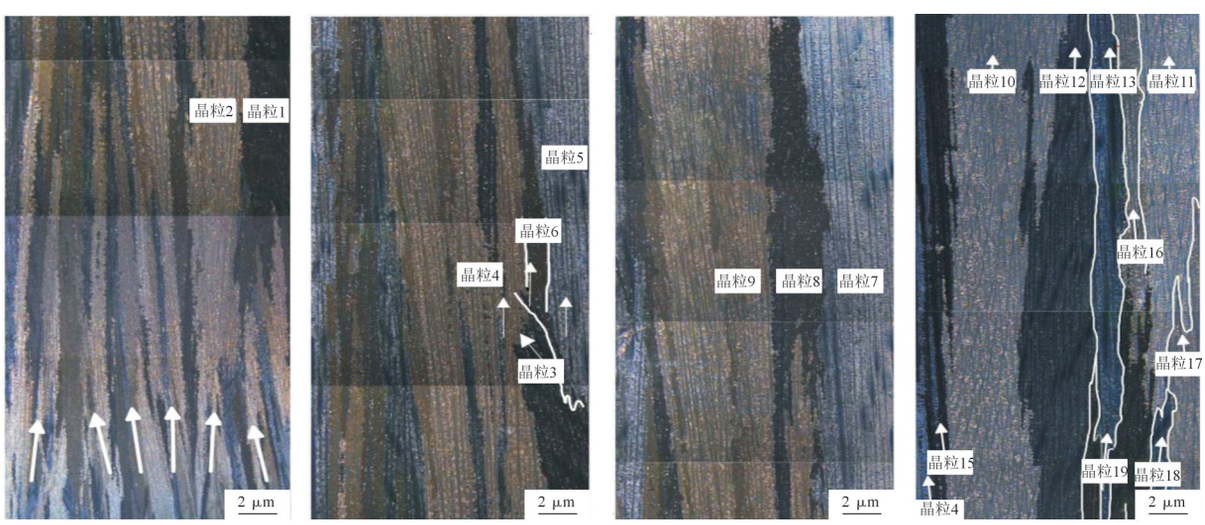
采用热力学计算软件 JMatProV6.2 中的 Scheil-Gulliver 模型，模拟计算 CM247LC 非平衡凝固过程随温度和液相含量变化的元素再分配规律和相生成规律。采用扫描电镜（SEM）附带能谱（EDS）以 3 mm 恒定距离测量铸态试样轴线上 60 个微区的元素组成，每个扫描区域面积为 1 500 μm（横向）× 500 μm（纵向）。

2　试验结果与分析

2.1　晶粒组织

2.1.1　试验结果

图 1 为 CM247LC 合金定向凝固铸件纵截面的腐蚀后体视显微组织。铸件下部（近水冷盘，凝固始端，每段长 30 mm）的纵截面宏观晶粒图像，清晰展现了定向凝固初始阶段具备多种取向的柱状晶剧烈竞争生长的过程，见图 1a～图 1c。铸件上部（远水冷盘，凝固末端，长 30 mm）纵截面的宏观晶粒图像存在的柱状晶的＜100＞择优取向与热流方向偏差均较小，见图 1d。凝



(a) 0~30 mm　(b) 30~60 mm　(c) 60~120 mm　(d) 120~180 mm

图 1　定向凝固试棒各个高度的纵截面晶粒形貌

Fig.1　Longitudinal grain morphologies of directionally solidified specimens at various heights

固末端得以保留的晶粒数目较少，且晶粒边缘犬牙交错，呈现非规则变化，由部分构成一个个完整封闭轨迹边界的晶粒，见图1d中的晶粒10、11及16、17，这是二维剖面不能完全展现三维特征所造成的。

图2为试棒纵截面上柱状晶粒数目及尺寸变化。可以看出，纵截面上的柱状晶数目在1～60 mm高度阶段迅速下降，而在此后的生长过程中变化减缓。对比图2中的柱状晶晶粒平均宽度和中位数宽度变化曲线则可以发现，1～30 mm高度上柱状晶晶粒平均宽度和中位数宽度相差不大，这说明在晶体生长的初始阶段柱状晶尺寸较为均匀。而在30 mm高度以上，竞争生长作用下柱状晶尺寸不一，晶粒宽度的中位数低于平均值，宏观晶粒组成为个别大尺寸晶粒与其余小尺寸晶粒。柱状晶生长至60 mm高度以上，直至凝固结束，柱状晶晶粒数目和尺寸的变化程度则相对较低。总体来说，随着生长高度增加，主要的晶粒竞争过程在1～60 mm高度上进行，大部分晶粒被淘汰，晶粒数量显著减少。

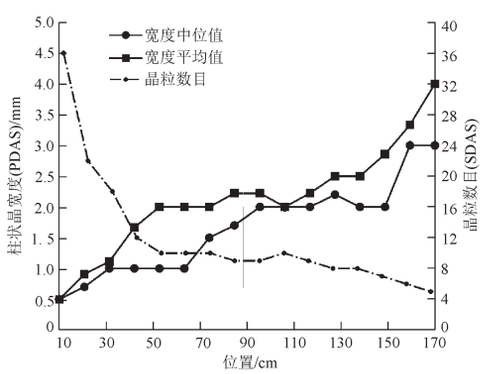


图2　试棒纵截面上柱状晶粒数目及尺寸变化

Fig.2　Variation trend of amounts and size of columnar grains along longitudinal section

2.1.2　分析与讨论

在图1a下部可以看到，由于水冷铜盘的激冷作用，凝固起始阶段的晶粒多而细密，这些不同衬度的柱状晶粒具备不同的择优取向并随机分布在不同的位置，随后沿择优取向快速生长。大多柱状晶的择优取向和热流的反方向具有一定的取向偏差角，取向偏差角较大的晶粒在竞争生长过程中会被那些择优取向和热流反方向一致的晶粒所淘汰，而择优取向沿热流反方向的晶粒得以继续生长。在图1a中晶粒1与晶粒2均得到了较好的发展，而在图1b中晶粒3与晶粒4呈汇聚生长，晶粒3与晶粒5呈发散生长，且最终晶粒1被淘汰，晶粒1与晶粒5交汇位置出现晶粒6。在距离水冷盘更远位置上，图1c中，具备较好择优取向的晶粒7～9占据了铸件剖面的绝大多数空间。

晶粒竞争生长的本质是枝晶分支的形成和晶界的移动，取决于晶粒间的相对取向及扩散场[7-8]。两晶粒发生交汇时，枝晶尖端溶质场重叠使边界层增大，元素在固液界面迁徙速度变慢[9-10]。图1b中，晶粒4相较于晶粒3与热流的取向偏差角小，生长领先，晶粒4扩散场的影响使晶粒3向左侧的生长受到抑制，同时晶粒4继续生长，其二次枝晶臂上形成3次枝晶臂并进一步发展形成平行于一次枝晶的枝晶干，从而使晶界向右发生了移动。也有研究指出，竞争生长过程中二次枝晶臂的取向对竞争不具备影响[17]。两相邻枝晶由于生长方向不同而逐渐分离时，发散晶界处存在不受溶质场影响的空白区域[15]。图1b中，晶粒4与晶粒因取向偏差大而呈发散关系，此时的晶粒6可能为新产生的晶粒，也可能是三维空间下从背侧生长过来的其他枝晶。

图1d处于凝固末期，此阶段截面晶粒组织数目更少，枝晶的择优取向和热流方向偏差角均较小，图1d所包含的30 mm长度内，仅淘汰了晶粒18和晶粒19，晶粒10～17得以保留。这一阶段晶粒的扩张与压缩仍受热流方向影响，也具有随机性。铸件在一组多件、环状排列的模组中进行定向凝固时，铸件朝内侧难以进行直接的辐射传热，且铸件模组的定向凝固过程中，各种凝固参数随时空变化，热流方向并不完全平行于提拉方向，这可能是晶粒14、晶粒15面积较小而能一直保留下来的原因[18]。

2.2　枝晶组织

2.2.1　试验结果

图3为定向凝固CM247LC合金的横向OM显微组织。可以看出，横截面枝晶呈二维十字轴枝晶形态。下部的枝晶较细小，上部较粗大。一定区域内，排列方式规律化，晶粒取向小范围内相同的十字轴枝晶构成柱状晶晶粒。距水冷盘3 mm高度上一次枝晶组织极其细密，见图3a，而在30 mm高度以上则不具备明显差异，尺寸较为均匀。

图4为定向凝固CM247LC合金的纵向OM显微组织。纵截面枝晶呈一定角度的连续、并列生长形态，二次枝晶的取向差异导致了二维纵截面枝晶臂粗细不一。结合图3与图4可以发现取向相同和取向不同的枝晶间均存在竞争生长，且自60 mm高度就出现了3次枝晶臂，这与竞争生长的晶界移动依靠枝晶分支的理论一致。

在试棒不同高度的横截面与纵截面各截取3个1.5 mm×1.5 mm的区域，根据式(1)与式(2)测出相应的一次枝晶臂间距，结果见图5。结合图3～图5可知，距离水冷盘较近处，枝晶组织相对致密，一次枝晶臂间距与二次枝晶臂间距均明显偏小；随着距水冷盘的高

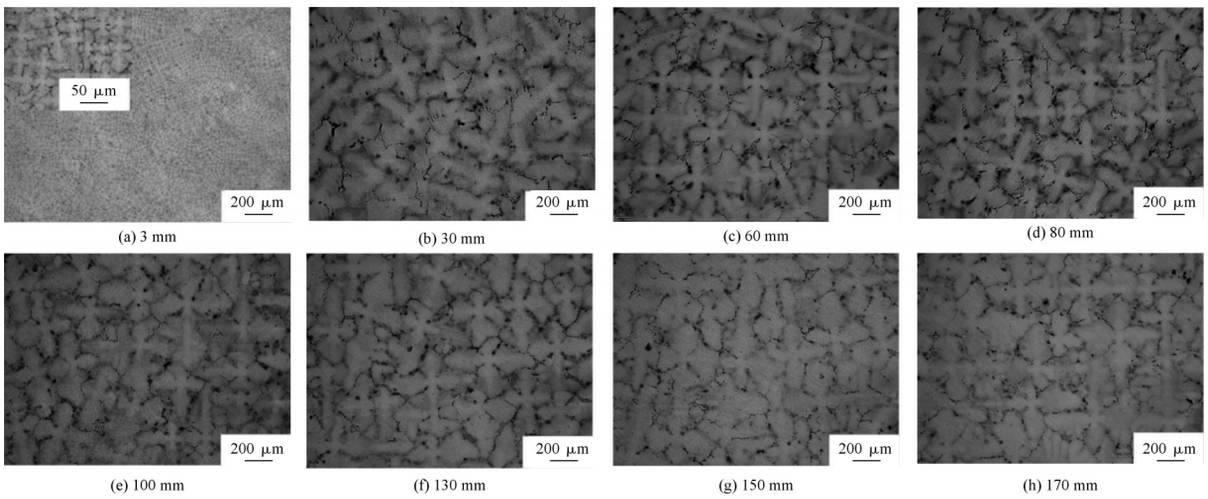


图 3　试棒不同高度处横截面的枝晶形貌

Fig.3　Cross-sectional dendritic morphologies of specimens at different heights

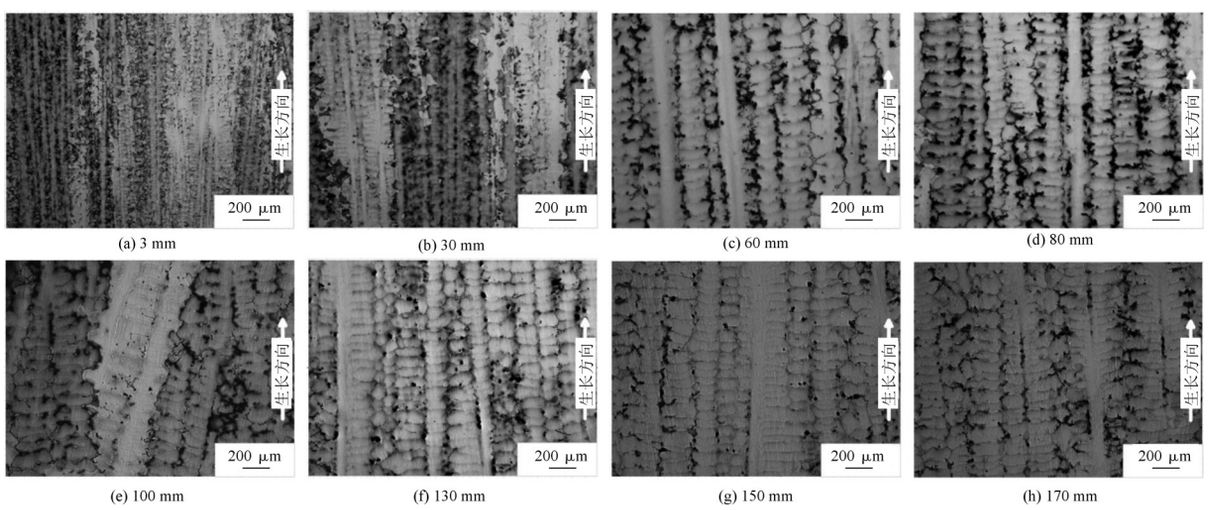


图 4　试棒不同高度处纵截面的枝晶形貌

Fig.4　Longitudinal dendritic morphologies of specimens at different heights

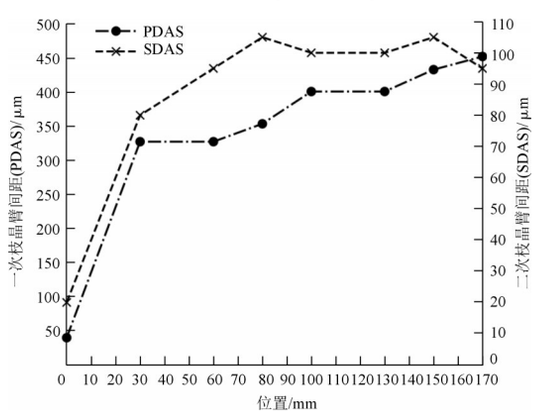


图 5　距水冷盘不同高度处的枝晶臂间距

Fig.5　Variaiton of dendrite arm spacing with distance from water-cooling plate

度提升，枝晶臂间距在 0～30 mm 高度上迅速增大，在 30～60 mm 高度上略有提升，在 60 mm 及以上的高度不再发生明显变化，一次枝晶间距在 450 μm 左右波动，二次枝晶间距在 100 μm 左右波动。

从图 3c 可以看出，60 mm 高度上即发生了二次枝晶臂表面失稳，出现了 3 次枝晶臂。随着枝晶生长过程推进，竞争生长作用下晶粒数目大幅下降，3 次枝晶所占比例逐渐提升，图 3f 中，出现 3 次分支的枝晶已占据截面面积的 50%。

2.2.2　分析与讨论

KURZ W 等[19]利用临界稳定性理论模型将一次枝晶间距（λ_1）与定向凝固过程中温度梯度（G）和生长速率（v）关系表述为：

$$\lambda_1 \propto G^{-1/2} \cdot v^{-1/4} \tag{3}$$

基于二次枝晶粗化假设的 FURER-WUNDERLIN 模型[20]将二次枝晶间距与定向凝固过程中温度梯度和生长速率关系表述为：

$$\lambda_2 \propto (G \cdot v)^{-1/3} \tag{4}$$

由式（3）和式（4）可知，随温度梯度和生长速度的降低，一次和二次枝晶臂间距均会增大。由图 3 和图 4

可知，随着距水冷盘高度的增加，试棒组织的一次枝晶间距和二次枝晶间距均明显增加。

通过数值模拟可得到试样轴线上间隔为5 mm的36个等距点的冷却曲线，经数据处理得到凝固前沿温度梯度G_i（$i\in[1,36]$）和柱状晶生长速度v_i的模拟值。将第i点的凝固前沿温度梯度G_i定义为该点冷却至液相线温度时，第$i+1$点与第i点温度差值与两点间距5 mm的比值：

$$G_i\approx\frac{\Delta T}{\Delta Z}=\frac{T_{i+1}-T_i}{5} \tag{6}$$

式中，T_{i+1}为$T_i=T_L$时第$i+1$个点的温度值，℃。

将柱状晶生长速度v定义为液相线移动速度，第i点位置生长速度的定义为：

$$v_i\approx\frac{\Delta Z}{\Delta t}=\frac{5}{t_{i+1}-t_i} \tag{7}$$

式中，t_i为$T_i=T_L$的时间节点，min；t_{i+1}为$T_{i+1}=T_L$的时间节点，min。

图6为试棒中凝固参数的模拟结果。对比图5和图6b可以发现，一次枝晶间距变化趋势与$G^{-1/2}\cdot v^{-1/4}$变化趋势基本保持一致，二次枝晶间距变化趋势与$(G\cdot R)^{-1/3}$亦高度正相关。这说明界面前沿液相的温度梯度和生长速率与定向凝固铸件的枝晶间距具有对应关系。以Bridgman法进行定向凝固时，固液界面逐渐远离水冷盘，其冷却作用减弱/导热效果变差，基本依赖型壳的辐射来散热，而固液界面前沿温度梯度相对降低，因而出现了图3、图4中枝晶间距先逐渐提升，后基本保持不变的规律。

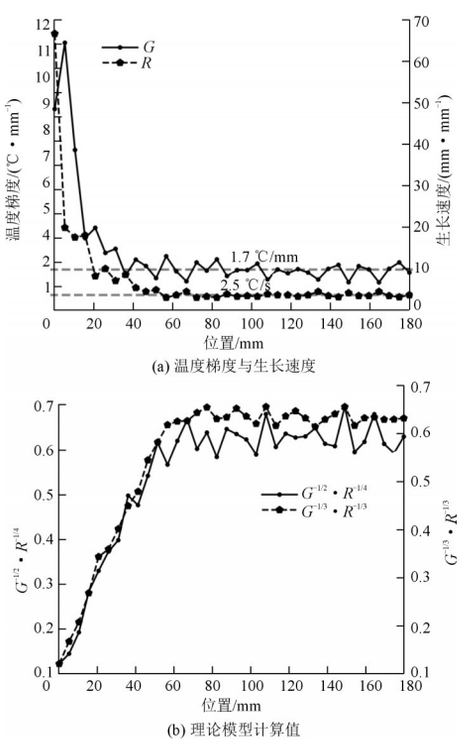


图6　试棒中凝固参数的模拟结果

Fig.6　Simulation results of solidification parameters for specimens

2.3　微观组织

2.3.1　试验结果

现有研究认为，共晶相和碳化物易在枝晶间析出，少见于枝晶干，这是由成分微观偏析和枝晶生长特性所决定的[24]。图7为CM247LC合金微观组织。可以看出，合金铸态组织中含有一定量的γ/γ'共晶和碳化物。γ/γ'共晶组织呈放射形态，周围常伴有缩孔存在。随着生长高度逐渐提升，存在着共晶组织逐渐增多和增大的迹象。图7a中共晶的尺寸明显小于后续高度上的共晶尺寸，且共晶团由图7b～图7d中的典型半圆形发展至图7e、图7f中的圆形。由于CM247LC中添加了少量C元素，所以其铸态组织不仅包括γ相枝晶和γ/γ'共晶团，还多出了初生碳化物MC相。图7中，碳化物形态为草书状，在10 mm高度上碳化物分布较为分散，30 mm高度以上的碳化物则主要呈现为连续的草书状，尺寸不存在明显变化。

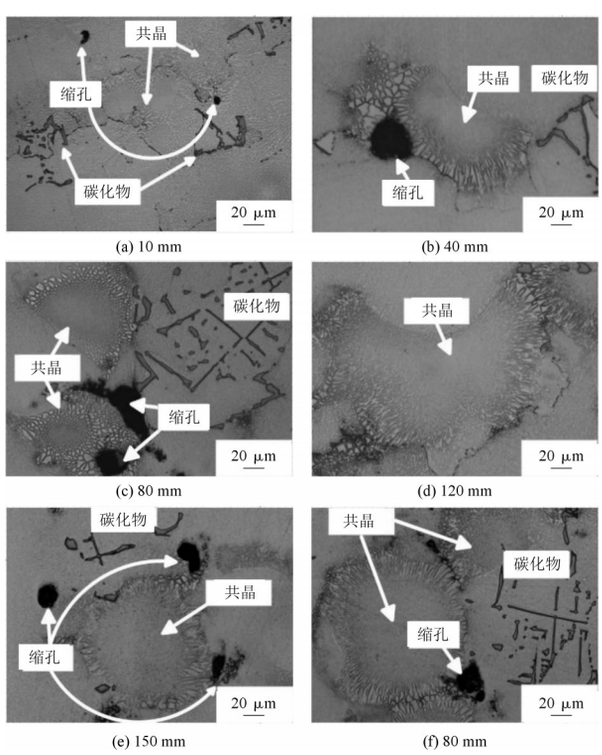


图7　试棒不同高度处的枝晶间微观组织

Fig.7　Interdendritic microstructure at different heights

缩松则主要分布在共晶边缘及枝晶间，主要呈圆形、不规则的长条形或三角形。一般认为，缩松的形成与镍基高温合金制备时界面前沿液相的温度梯度和抽拉速率相关，随着抽拉速率增加，缩松孔径会不断减

小。试验结果表明，仅在 0～40 mm 的凝固初始端，缩孔随着距水冷盘高度的增加而增加，在距水冷盘更大距离的范围内，缩孔尺寸变化不大。

2.3.2　分析与讨论

图 8 为 CM247LC 合金热力学计算结果。由图 8a 及文献[20-21]可知，镍基高温合金 CM247LC 在凝固时首先从液相中析出 γ 固溶体，随后在枝晶间区域形成 *MC* 碳化物；1 255 ℃时，凝固过程接近结束时，固相分数约为 95%，发生共晶反应：$L_3 \rightarrow \gamma/\gamma'$ 及 $L_3+\gamma \rightarrow \gamma/\gamma'$。1 210 ℃时，固相分数约为 98%，随着温度继续降低，γ 固溶体中的溶质原子溶解度降低，过饱和 γ 固溶体发生固态相变，γ′ 相开始形核析出并长大。

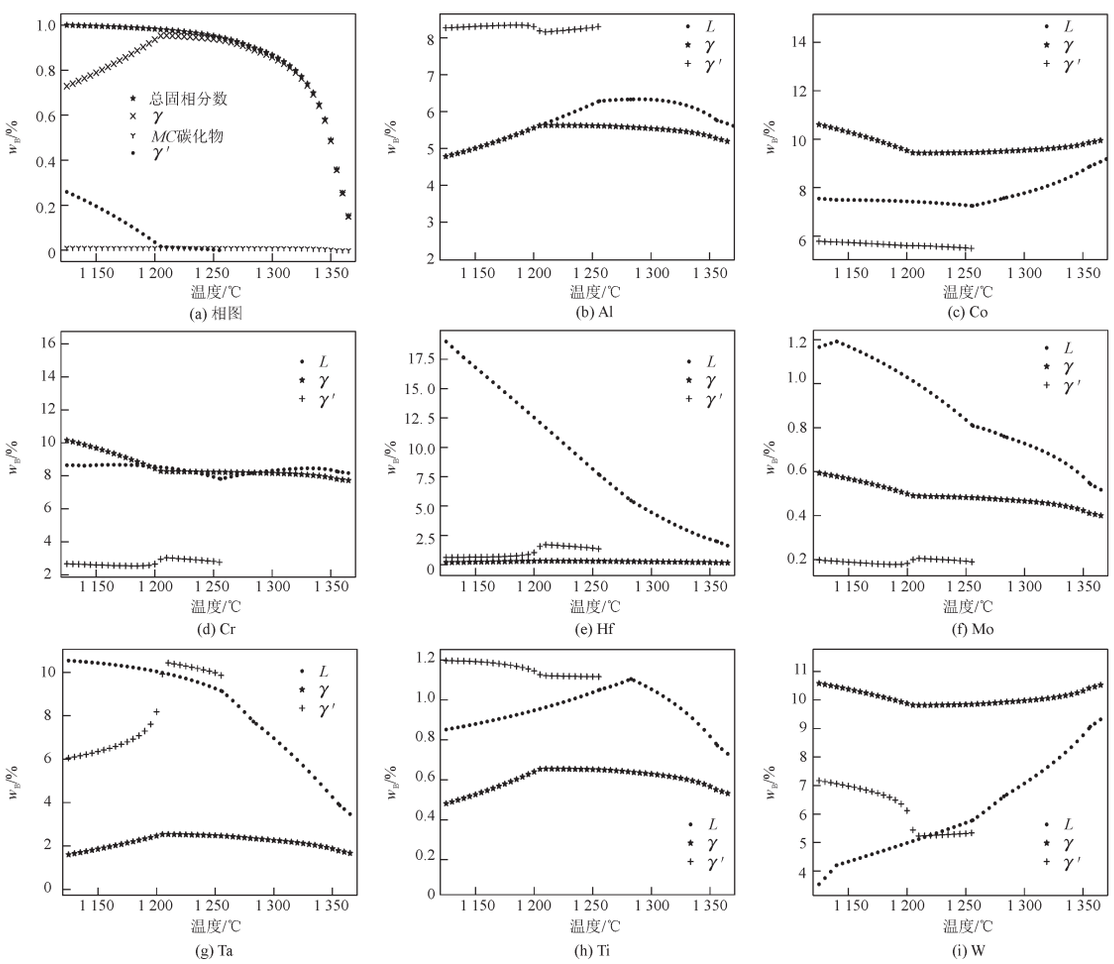


图 8　CM247LC 合金热力学计算结果

Fig.8　Thermodynamic calculation results of CM247LC alloy

γ/γ′ 共晶团的核心在瞬间通过共晶反应发生，紧接着是 γ/γ′ 的快速生长，逐渐过渡至粗大的层片状共晶，所以核心部位的共晶组织非常细密，周边组织则呈粗大发散状，形成花瓣状的组织，且有研究表明自共晶心部到周边 Co、Cr、W 元素含量逐渐下降，Al、Hf、Ta、Ti 含量逐渐提高[27]。这一成分变化与图 8b～图 8i，凝固过程中共晶温度区间（1 210～1 255 ℃，此温度区间内生成的 γ 相成分不存在明显变化）内生成的 γ′ 相成分的变化基本一致，说明热力学计算结果可信。

现有研究认为含碳镍基高温合金共晶团具有几种典型模式，分别为在 γ 相上形成 γ 基共晶、在 *MC* 碳化物上形核生成和从液体中形核形成自由共晶[23-24]。其中，以 γ 相为基底形成的共晶体为近似半圆形，其共晶核心处在 γ 枝晶端部，具有与 γ 相基底相同的晶体取向；以 MC 碳化物作为 γ/γ′ 共晶反应核心的共晶团晶体取向由碳化物决定；从液体中自由形核并长大的 γ/γ′ 共晶团近似为圆形，共晶核心处在圆的中心，其晶体取向是随机的[28]。共晶的生长一直持续到与其他共晶团或邻近的 γ 枝晶接触，所以 γ 枝晶的尺寸与间距限制了共晶的尺寸。共晶组织尺寸大小不一，形状各异。离水冷盘越远，共晶组织尺寸越大，是因为凝固开始阶段的一次枝晶间距小，组织致密，其间距限制了共晶组织的长大，因而共晶组织尺寸变小。

高温合金的定向凝固过程为自下而上的顺序凝固，因此宏观缩松较为少见，多数表现为枝晶间的显微缩松。随着距离水冷铜盘越来越大，枝晶臂间距不断

增大，见图5和图6，枝晶间的熔池尺寸也相对增大，其凝固的组织也越来越不均匀，共晶尺寸逐渐增加，缩孔尺寸也明显增大。这是因为定向凝固过程中最后凝固的区域为枝晶间区域，糊状区的枝晶之间互相交叉，将液态金属分割为数量较多的小熔池，当金属液体被已生成的枝晶臂所隔离而得不到外来补缩时，会形成枝晶间显微缩松。有研究认为，含碳镍基高温合金在定向凝固过程中，糊状区中碳化物的存在会显著降低糊状区渗透性而阻隔液相补缩，从而加剧缩孔[25-26]。这与图7中观测到的共晶处缩孔与碳化物相伴出现这一特征相符。

定向凝固中元素的偏析主要为枝晶干与枝晶间存在的微观偏析，宏观尺度上成分偏析不明显，研究镍基高温合金的微观元素偏析非常重要[32]。图9为定向凝固试棒纵截面上的成分分布测量结果，表1为各元素在高温合金中的偏析系数。由图9可以看出，宏观区域成分测量结果也验证了镍基高温合金试样在定向凝固高度方向上不存在明显的宏观偏析。结合图8与表1可知，随着液相体积分数降低，Al、Ti、Ta、Hf等正偏析元素（$k<1$，图8b、图8e、图8g、图8h）在液相中的浓度增加，因此主要富集于枝晶间，最终达到共晶点，形成图7所示的γ/γ'共晶相；同时W、Co等负偏析元素（$k<1$，图8c、图8i）含量降低，则偏聚于枝晶干，这与现有的试验测量结果基本一致[28-32]。这种枝晶生长过程中，各元素偏析系数差异导致的微观偏析，对铸件的宏观成分分布影响较小，但在微观尺度会造成成分的剧烈波动，即枝晶偏析。

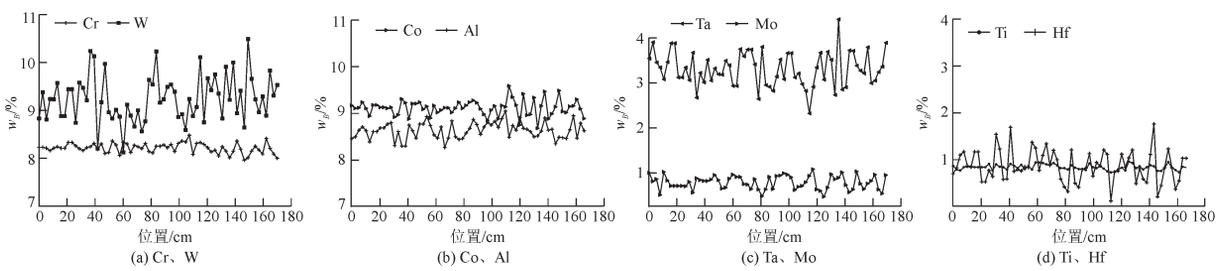


图9　定向凝固试棒纵截面上的成分分布测量结果

Fig.9　Measured component distribution of directionally solidified specimen along longitudinal section

表1　各元素在高温合金中的偏析系数

Tab.1　Segregation coefficient of each element in superalloy

来源	偏板系数							
	Al	Ti	Cr	Co	Mo	Hf	Ta	W
文献[39]	0.91	-	1.05	1.06	-	-	0.76	1.47
文献[36]	0.91	0.88	0.97	1.12	0.86	-	0.54	-
相图计算	0.89	0.67	1.04	1.27	0.62	0.18	0.35	1.64

图10为轴向元素分布标准差。值得注意的是，尽管偏聚方向不同，宏观尺度生长方向上元素成分的波动程度与其原子序数近似呈正相关，这可能是溶质元素在γ相中固溶度大小所决定的，也可能是因为原子半径较大的原子在液相中扩散系数相对较低，在液相中混合均匀困难所造成的[38]。

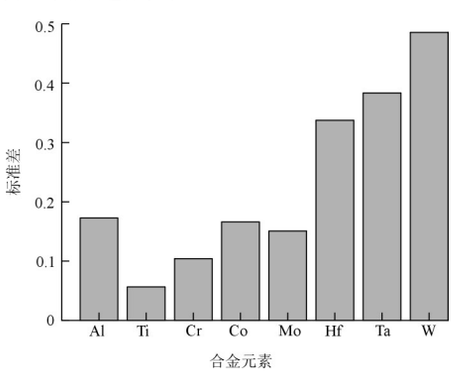


图10　轴向元素分布标准差

Fig 10　Standard deviation of axial element distribution

3　结　论

（1）在高温合金CM247LC的定向凝固试样中，主要的柱状晶粒竞争过程发生在1～60 mm高度上，这一阶段内大部分晶粒被淘汰，晶粒数量大大减小；在60 mm以上的高度上，晶粒数目减少和晶粒尺寸增加的趋势明显减缓。

（2）CM247LC的铸态组织主要呈现为γ树枝晶结构，在凝固起始阶段枝晶间距迅速增加，在60 mm高度以上逐渐减缓。枝晶臂间距的变化趋势与温度场模拟结果具有良好的对应关系。

（3）CM247LC的铸态组织中含有一定量的γ/γ'共晶和MC碳化物，还存在明显的微观缩孔。共晶、碳化物和微观缩孔的尺寸变化与枝晶臂间距的变化趋势基本一致。

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